

Instructions:

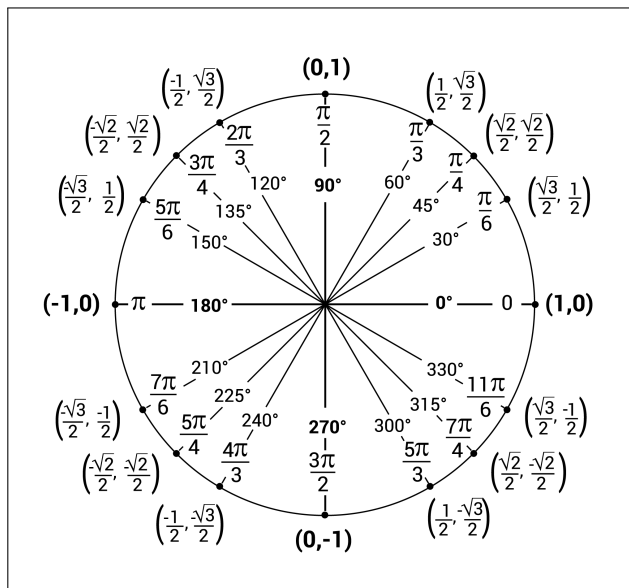
- Make sure to clearly show each step of your solution for full credit - correct answers without any work will not receive credit. Some questions may remind you to show your work explicitly, but you should assume each question requires you to justify your steps by making your thought process clear in your solution. You can use the solutions we write in class as an indication of how much work you are expected to show. You can ask the instructor if it is not clear how much work you are expected to show on a particular problem.
- Write all of your work and answers on this document. If you need extra paper, please ask the instructor. If a box is provided for your final answer, please write your final answer in the box.
- Indicate your affirmation of the following statement by signing below.

I affirm that I did not give or receive any unauthorized assistance to solve the problems below. I understand that the penalty for academic dishonesty is an F for the course and possible suspension or expulsion from the university.

Signature: _____

Formulas

- $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1, \lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$
- Continuity at $x = a$: $\lim_{x \rightarrow a} f(x) = f(a)$
- $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
- Basic derivatives
 - $\frac{d}{dx} (c \cdot f(x)) = c \cdot f'(x)$
 - $\frac{d}{dx} (f(x) \pm g(x)) = f'(x) \pm g'(x)$
 - $\frac{d}{dx} (x^n) = nx^{n-1}$
 - $\frac{d}{dx} (e^x) = e^x$
 - $\frac{d}{dx} (\ln(x)) = \frac{1}{x}$
 - $\frac{d}{dx} (\sin x) = \cos x$
 - $\frac{d}{dx} (\cos x) = -\sin x$
 - $\frac{d}{dx} (\tan x) = \sec^2 x$
 - $\frac{d}{dx} (\sec x) = \sec x \cdot \tan x$
 - $\frac{d}{dx} (\cot x) = -\csc^2 x$
 - $\frac{d}{dx} (\csc x) = -\csc x \cdot \cot x$
- Derivative rules
 - $\frac{d}{dx} (f(x) \cdot g(x)) = f'(x) \cdot g(x) + g'(x) \cdot f(x)$
 - $\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}$
 - $\frac{d}{dx} (f(g(x))) = f'(g(x)) \cdot g'(x)$
- $L(x) = f'(a)(x - a) + f(a)$



A1 - Limits and continuity (inspired by parts 2 of the class notes and Brightspace problems 4 and 10)

(a) Evaluate $\lim_{x \rightarrow -\infty} \sqrt{x^6 - 4x^3 + x^2 + x + 1} + x^3$.

$$\begin{aligned}
 &= \lim_{x \rightarrow \infty} \frac{\sqrt{x^6 + 4x^3 + x^2 - x + 1} - x^3}{1} \cdot \frac{\sqrt{x^6 + 4x^3 + x^2 - x + 1} + x^3}{\sqrt{x^6 + 4x^3 + x^2 - x + 1} + x^3} = \lim_{x \rightarrow \infty} \frac{\cancel{x^6} + 4x^3 + x^2 - x + 1 - \cancel{x^6}}{\sqrt{\cancel{x^6} + 4x^3 + x^2 - x + 1} + x^3} \cdot \frac{\frac{1}{x^3}}{\frac{1}{x^3}} \\
 &= \lim_{x \rightarrow \infty} \frac{4 + \frac{1}{x} - \frac{1}{x^2} + \frac{1}{x^3}}{\sqrt{1 + \frac{4}{x^3} + \frac{1}{x^4} - \frac{1}{x^5} + \frac{1}{x^6}} + 1} = \frac{4}{2} = \boxed{2}
 \end{aligned}$$

(b) Evaluate $\lim_{x \rightarrow 1^-} \frac{|x^2 - 1|}{x^2 - 3x + 2}$.

$$\begin{aligned}
 &= \lim_{x \rightarrow 1^-} \frac{|(x-1)(x+1)|}{(x-1)(x-2)} = \lim_{x \rightarrow 1^-} \frac{\cancel{(x-1)}(x+1)}{\cancel{(x-1)}(x-2)} = \frac{-(1+1)}{(1-2)} = \boxed{2}
 \end{aligned}$$

(c) Show that there are at least two x -values where the graphs of $f(x) = 3^{1-x^2} - 1$ and $g(x) = 2 - \cos(x)$ intersect. Be sure to fully justify your reasoning.

Both $f(x)$ and $g(x)$ are continuous (compositions/differences of polynomial/exponential/trigonometric functions), so $h(x) = f(x) - g(x)$ is continuous. $f(x)$ and $g(x)$ intersect where $h(x) = 0$.

$$h(-1) = 3^{1-(-1)^2} - 1 - (2 - \cos(-1)) = 3^0 - 1 - 2 + \cos(-1) = -2 + \cos(-1) < 0$$

$$h(0) = 3^{1-0^2} - 1 - (2 - \cos(0)) = 3 - 1 - 2 + \cos(0) = 1 > 0$$

$$h(1) = 3^{1-1^2} - 1 - (2 - \cos(1)) = 3^0 - 1 - 2 + \cos(1) = -2 + \cos(1) < 0$$

By IVT, there is a c_1 in $(-1, 0)$ and a c_2 in $(0, 1)$

such that $h(c_1) = 0$ and $h(c_2) = 0$.

A2 - The derivative (inspired by part 2 of the class notes and past final worksheet problem 3)

- (a) Use the limit definition of the derivative to find the derivative of $f(x) = \frac{-3}{x^2}$. You must use the limit definition of the derivative to get any credit for this problem.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} &= \lim_{h \rightarrow 0} \frac{\frac{-3}{(x+h)^2} - \frac{-3}{x^2}}{h} \cdot \frac{x^2(x+h)^2}{x^2(x+h)^2} = \lim_{h \rightarrow 0} \frac{-3x^2 + 3(x+h)^2}{h \cdot x^2 \cdot (x+h)^2} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{-3x^2} + \cancel{3x^2} + 6xh + 3h^2}{h \cdot x^2 \cdot (x+h)^2} = \lim_{h \rightarrow 0} \frac{\cancel{6xh} + 3h^2}{\cancel{h} x^2 (x+h)^2} = \lim_{h \rightarrow 0} \frac{6x + 3h}{x^2 (x+h)^2} = \frac{6x}{x^2 \cdot x^2} = \boxed{\frac{6}{x^3}} \end{aligned}$$

- (b) Consider the piecewise function

$$h(x) = \begin{cases} x+1 & x \leq 1 \\ 2\sqrt{x} & x > 1 \end{cases}$$

Use the limit definition of the derivative to determine if $h(x)$ is differentiable at $x = 1$. You must use the limit definition of the derivative to get any credit for this problem.

$$\begin{aligned} \lim_{x \rightarrow 1^-} \frac{h(x) - h(1)}{x - 1} &= \lim_{x \rightarrow 1^-} \frac{x+1-2}{x-1} = \lim_{x \rightarrow 1^-} \frac{x-1}{x-1} = 1 \\ \lim_{x \rightarrow 1^+} \frac{h(x) - h(1)}{x - 1} &= \lim_{x \rightarrow 1^+} \frac{2\sqrt{x} - 2}{x - 1} \cdot \frac{2\sqrt{x} + 2}{2\sqrt{x} + 2} = \lim_{x \rightarrow 1^+} \frac{4x - 4}{(x-1)(2\sqrt{x} + 2)} \\ &= \lim_{x \rightarrow 1^+} \frac{\cancel{4(x-1)}}{\cancel{(x-1)}(2\sqrt{x} + 2)} \\ &= \lim_{x \rightarrow 1^+} \frac{4}{2\sqrt{x} + 2} = \frac{4}{2\sqrt{1} + 2} = 1 \end{aligned}$$

So yes, $h'(1)$ exists and is equal to 1.

A3 - Derivative rules (inspired by Brightspace problem 5 and past final worksheet problem 4)

- (a) Find the derivative of $f(x) = x^{\cos(\ln(x))}$. You do not need to simplify your answer.

$$\ln(f(x)) = \ln(x^{\cos(\ln(x))}) = \cos(\ln(x)) \cdot \ln(x)$$

$$\frac{f'(x)}{f(x)} = -\sin(\ln(x)) \cdot \frac{1}{x} \cdot \ln(x) + \cos(\ln(x)) \cdot \frac{1}{x}$$

$$f'(x) = x^{\cos(\ln(x))} \cdot \left[-\sin(\ln(x)) \cdot \frac{1}{x} \cdot \ln(x) + \cos(\ln(x)) \cdot \frac{1}{x} \right]$$

- (b) Find the derivative of $g(x) = \sqrt[5]{\frac{e^x + a}{2x^3 + ax}}$ where a is a constant. You do not need to simplify your answer.

$$g'(x) = \frac{1}{5} \left(\frac{e^x + a}{2x^3 + ax} \right)^{-\frac{4}{5}} \cdot \left(\frac{(2x^3 + ax) \cdot e^x - (e^x + a) \cdot (6x^2 + a)}{(2x^3 + ax)^2} \right)$$

A4 - Implicit curves and linearization (inspired by part 2 of the class notes and Brightspace problem 3)

- (a) Find the slope of the implicit curve $\tan(xy) = \frac{y}{1+x^2} - \frac{\pi}{6} + \sqrt{3}$ at $(1, \frac{\pi}{3})$. You do not need to simplify your answer.

$$\sec^2(xy) \cdot (1+y+x \cdot \frac{dy}{dx}) = \frac{(1+x^2) \cdot \frac{dy}{dx} - y \cdot 2x}{(1+x^2)^2}$$

$$\sec^2(1 \cdot \frac{\pi}{3}) \cdot (\frac{\pi}{3} + \frac{dy}{dx}) = \frac{2 \frac{dy}{dx} - \frac{2\pi}{3}}{4}$$

$$4 \cdot (\frac{\pi}{3} + \frac{dy}{dx}) = \frac{\frac{dy}{dx} - \frac{\pi}{3}}{2}$$

$$\frac{8\pi}{3} + 8 \frac{dy}{dx} = \frac{dy}{dx} - \frac{\pi}{3}$$

$$3\pi = -7 \frac{dy}{dx} \rightarrow \frac{dy}{dx} = \frac{3\pi}{-7}$$

- (b) Use linearization to approximate $\frac{1}{\sqrt[3]{7.9}}$. You do not need to simplify your answer.

$$f(x) = \frac{1}{\sqrt[3]{x}} = x^{-\frac{1}{3}} \quad @ \quad a=8$$

$$f'(x) = -\frac{1}{3} x^{-\frac{4}{3}}$$

$$f(8) = \frac{1}{\sqrt[3]{8}} = \frac{1}{2}$$

$$f'(8) = -\frac{1}{3} (8)^{-\frac{4}{3}} = -\frac{1}{48}$$

$$L(x) = -\frac{1}{48} (x-8) + \frac{1}{2}$$

$$\frac{1}{\sqrt[3]{7.9}} = f(7.9) \approx L(7.9) = -\frac{1}{48} \cdot (7.9-8) + \frac{1}{2}$$

$$= \frac{1}{480} + \frac{1}{2}$$

$$= \frac{241}{480}$$

A5 - Related rates (inspired by Brightspace problems 2 and 7 and the class notes)

- (a) A sphere is shrinking so that its volume is decreasing at a rate of $16 \text{ cm}^3/\text{min}$. How fast is the sphere's radius decreasing when the volume is $\frac{32\pi}{3} \text{ cm}^3$? Include units in your answer. *Hint:* The volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$.

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dt} = -16$$

$$\frac{dV}{dt} = 4\pi r^2 \cdot \frac{dr}{dt}$$

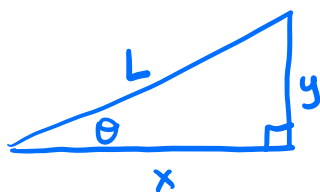
$$V = \frac{32\pi}{3} \rightarrow \frac{32\pi}{3} = \frac{4}{3}\pi r^3$$

$$\rightarrow r = 2$$

$$-16 = 4 \cdot \pi \cdot 2^2 \cdot \frac{dr}{dt}$$

$$\boxed{\frac{dr}{dt} = -\frac{1}{\pi} \text{ cm/min}}$$

- (b) A person is flying a kite with a constant length of string, and the wind begins to cause the kite's height to decrease at a constant rate of 2 feet per second. At what rate is the kite's angle of elevation changing when the horizontal distance from the person to the point on the ground directly below the kite is 100 feet. Include units in your answer.



$$L \text{ is constant so } \frac{dL}{dt} = 0$$

$$\frac{dy}{dt} = -2$$

$$\cos \theta = \frac{x}{L}$$

$$\sin \theta = \frac{y}{L}$$

$$\cos \theta \cdot \frac{d\theta}{dt} = \frac{L \cdot \frac{dx}{dt} - y \cdot \frac{dL}{dt}}{L^2} = \frac{L \cdot \frac{dx}{dt}}{L^2} = \frac{-2}{L}$$

$$\frac{x}{L} \cdot \frac{d\theta}{dt} = \frac{-2}{L} \rightarrow \frac{d\theta}{dt} = \frac{-2}{x} = \frac{-2}{100} = \boxed{-0.02 \text{ rad/s}}$$